FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XII

JEE (Main)-2022

TEST DATE: 22-05-2022

ANSWERS, HINTS & SOLUTIONS

Physics PART – A

SECTION – A

1. A

Sol. As, $B = \mu_0(H+I)$

$dB = \mu_0 dH + \mu_0 dI$

$\oint H dB = \mu_0 \oint H dH + \mu_0 \oint H dI$

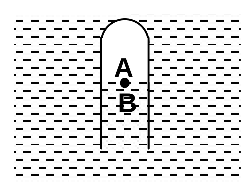
$A_2 = \mu_0 A_1$ ($\because \oint H dH = 0$)

2. B

Sol. beat frequency = $|f_1 - f_2|$

3. D

Sol. $P_A = P_B$



$P_A(L - h)A = P_0 LA$ ($\because P_1V_1 = P_2V_2$)

$\frac{P_0 L}{(L-h)} = P_0 + \rho g(L-h)$

Also, $P_0 = \rho g L$ ($\because$ given)

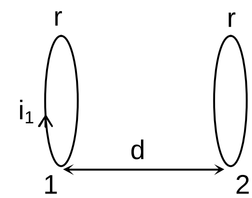
$(2L - h)(L - h) = L^2$

4. D

Sol. $\phi_2 = Mi_1$

$\frac{\mu_0 i_1 r^2}{2d^3}(\pi r^2) = Mi_1$

$\therefore\ M = \frac{\mu_0 \pi r^4}{2d^3}$



5. A

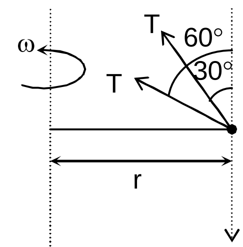
Sol. F.B.D. of bead is as

T cos 30° + T sin 30° = mg …(i)

T cos 60° + T sin 60° = $m\omega^2 r$ …(ii)

From (i) and (ii)

$\omega = \sqrt{\frac{g}{r}}$



6. C

Sol. Exponentially decreasing.

7. B

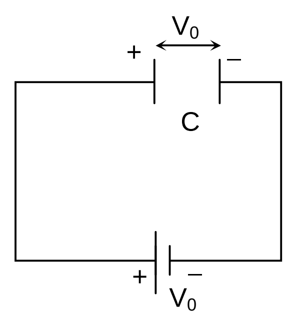
Sol. $\frac{dN}{dv} = 0$ for v_P (because N is maximum for $v = v_P$)

$v_{rms} > r_m > v_P$

8. D

Sol. After closing the switch, the circuit will look like.

H.L. = $W_b - \Delta U = V_0(2CV_0) - 0 = 2CV_0^2$ ($\Delta U = 0$)



9. D

Sol. $\phi_C + \phi_P = \frac{q}{\varepsilon_0}$ (Gauss law)

$\phi_C = \frac{q}{\varepsilon_0} - \frac{q}{2\varepsilon_0}\left(1 - \frac{1}{\sqrt{2}}\right)$

$\frac{\phi_C}{2} = \frac{q}{4\varepsilon_0}\left(1 + \frac{1}{\sqrt{2}}\right)$

10. C

Sol. Rotational equilibrium $\Rightarrow$ fR – Tf = 0

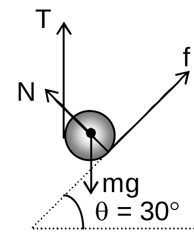
T = f …(i)

Translational equilibrium along the inclined plane

$\therefore$ T sin θ + f = mg sin θ …(ii)

From (i) and (ii)

$$T = \frac{mg\sin\theta}{1+\sin\theta} = \frac{mg}{3}$$



11. C

Sol. The system must be either capacitive or inductive and also non-resistive

12. A

Sol. $a = a_0 e^{-kt}$, $\frac{\ell n2}{k} = t_0$

$\Rightarrow a = a_0 e^{-(\ell n2)\frac{t}{t_0}}$

$$\int_0^{v_T} dv = \int_0^{\infty} a_0 e^{-\frac{\ell n2t}{t_0}} dt$$

$\Rightarrow v_T = \frac{a_0 t_0}{\ell n2}$

13. A

Sol. $f = RC(Z-b)^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$\sqrt{f}$ for k_β is $\sqrt{RC(Z-1)^2\left(1-\frac{1}{9}\right)}$

$\sqrt{f}$ for k_α is $\sqrt{RC(Z-1)^2\left(1-\frac{1}{4}\right)}$

Required ratio = $\sqrt{\frac{8RC}{9}}\sqrt{\frac{4}{3RC}} = \sqrt{\frac{32}{27}}$

14. A

Sol. As voltage across voltmeter is 20V, its resistance is ∞ and hence negligible current in voltmeter.

$\therefore$ iR = 20

$\frac{24R}{1+R} = 20$

24R = 20R + 20

4R = 20

R = 5Ω

15. B

Sol. Refraction at curved surface

$\frac{3}{2v_1} - \frac{1}{-\infty} = \frac{\frac{3}{2}-1}{+6}$, v_1 = +18 cm

Refraction at plane surface

$\frac{1}{v} = \frac{3}{2(+12)} \Rightarrow$ v = 8 cm

16. C

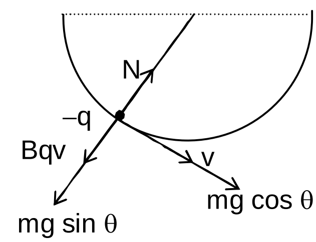
Sol. Applying work energy theorem, we get

V at Q is $\sqrt{2gR\sin\theta}$

$$N - mg\sin\theta - Bqv = \frac{mv^2}{R}$$

$$N = mg\sin\theta + Bq\sqrt{2gR\sin\theta} + 2mg\sin\theta$$

$$N = 3mg\sin\theta + Bq\sqrt{2gR\sin\theta}$$



17. D

Sol. Thin wire are made up of ductile materials.

18. C

Sol. $C = k\varepsilon_0 \frac{A}{d} = 8.85 \times 10^{-8} F$

$$i_d = \frac{d}{dt}(CV) = 2.2\mu A$$

19. B

Sol. On the experimental basis, k_2 should be pressed first.

20. B

Sol. no. of protons = 72 – 4 + 1 = 69

Mass number = 180 – 8 = 172

SECTION – B

21. 00003.00

Sol. $g \propto \rho R$ and $h \propto \frac{1}{g}$

$\therefore$ $h\rho R$ = same for both

$(0.5)\rho R = h\left(\frac{2}{3}\right)\rho\left(\frac{R}{4}\right) \Rightarrow h = 3m$

22. 00002.00

Sol. $v_{rms} = \sqrt{2}v_{sound}$

$$\sqrt{\frac{3RT}{M}} = \sqrt{2}\sqrt{\frac{Y_{mix}RT}{M}}$$

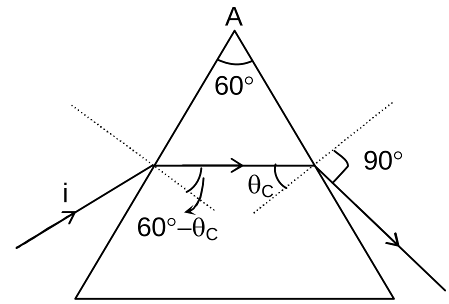
$$Y_{mix} = \frac{3}{2}$$

$$\frac{n+2}{\frac{3}{2}-1} = \frac{n}{\frac{7}{5}-1} + \frac{2}{\frac{5}{3}-1}$$

$\Rightarrow n = 2$

23. 00001.50

Sol. For limiting angle of incidence, the angle of emergence is 90°.



$\sin i = \mu \sin(60 - \theta_C)$

$\theta_C = \sin^{-1}\left(\sqrt{\frac{3}{7}}\right)$

$\therefore\ i = \frac{\pi}{6}$, 4k = 6 and k = 1.5

24. 00002.22

Sol. $Q_1 = 2\left(\frac{7R}{2}\right)20$, $Q_2 = 3\left(\frac{3R}{2}\right)14$

$\frac{Q_1}{Q_2} = 2.22$

25. 00001.33

Sol. $\frac{hC}{\lambda} - \phi = eV$

$\frac{3hC}{\lambda_0} - \frac{hC}{\lambda_0} = \frac{eq}{4\pi\varepsilon_0 R}$

$\Rightarrow\ q = \frac{8\pi\varepsilon_0 RCh}{e\lambda_0}$

$\therefore$ 8 = 6n

n = 4/3 = 1.33

26. 00002.00

Sol. $\frac{d\ell}{\ell} \times 100 = 1$

A = $6\ell^2$

$\frac{dA}{A} \times 100 = \frac{12\ell d\ell}{6\ell^2} \times 100 = 2$

27. 00000.17

Sol. k = 75% of E $\Rightarrow$ U = 25% of E

$\therefore$ x = A/2

The time taken from x = 0 to x = A/2 is $\frac{T}{12} = \frac{1}{6}$sec

28. 00000.20

Sol. $\frac{1}{2}L(4)^2 = 32$ L = 4H

$(4)^2$R = 320 R = 20Ω

$t_0 = \frac{L}{R} = 0.2$ sec

29. 00040.20

Sol. $\dfrac{t - t_{FP}}{t_{BP} - t_{FP}} = \text{constant}$

$\dfrac{t-1}{99-1} = \dfrac{40-0}{100-0} \quad t = 1 + 98\left(\dfrac{4}{10}\right)$

t = 40.2°C

30. 00000.20

Sol. $I = I_m \cos^2\left(\dfrac{\pi y d}{\lambda D}\right)$

$\cos^2\left(\dfrac{\pi y d}{\lambda D}\right) = \dfrac{3}{4}$

$|\Delta y|_{min} = 0.2$ mm

Chemistry

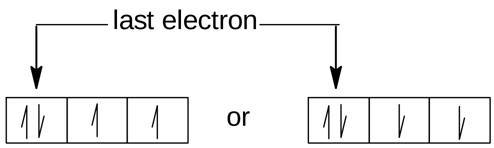
PART – B

SECTION – A

31. D

Sol. Electronic configuration of 'S'

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$



32. C

Sol. In case of P – Cl bond, orbital, of 3rd shell of Cl is involved in bonding while in case of P – F bond, orbital of 2nd shell of F is involved, which is much smaller than 3rd shell orbital. Also, axial bond length is larger than equitorial bond length, provided the ligand atoms are same.

33. C

Sol. $$K_C = \frac{K_P}{(RT)^2} = \frac{50}{(304.5 \times 0.0821)^2} = 0.08$$

$$\begin{array}{lcll} 2ClF_3 & \rightleftharpoons & Cl_2 & + \ 3F_2 \\ (0.4 - x) & & \frac{x}{2} & \ \ \frac{3x}{2} \end{array}$$

$$0.08 = \frac{\left(\frac{x}{2}\right) \times \left(\frac{3x}{2}\right)^3}{(0.4 - x)^2}$$

On solving x = 0.12915

$$[Cl_2] = \frac{x}{2} = 0.065\ M(\text{approx.})$$

34. A

Sol.
$$\begin{array}{lll} AgBr(s) \rightleftharpoons Ag^+(aq) + Br^-(aq);\ K_{SP} \\ \qquad\qquad S \qquad\qquad S \\ \qquad\qquad \sim \end{array}$$

$$\begin{array}{llll} Ag^+ + & 2NH_3(aq) \rightleftharpoons & [Ag(NH_3)_2]^+; & K_f \\ S & 0.1\ M & 0 & \\ \sim & 0.1 - 2S & S & \end{array}$$

$$\begin{array}{l} AgBr(s) + 2NH_3(aq) \rightleftharpoons Br^-(aq) + [Ag(NH_3)_2]^+(aq) \\ \qquad\quad (0.1 - 2S) \qquad\quad S \qquad\qquad S \end{array}$$

$$K_{SP} \times K_f = \frac{S^2}{(0.1 - 2S)^2}$$

$$5 \times 10^{-13} \times 10^{+7} = \left(\frac{S}{0.1 - 2S}\right)^2$$

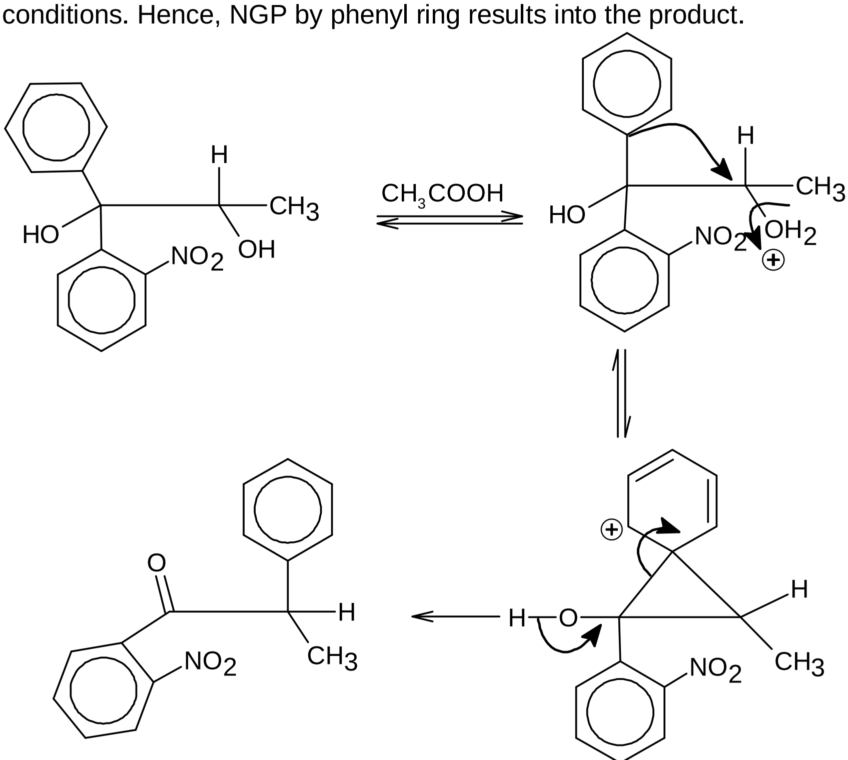
$$S = 2.226 \times 10^{-4}\ M$$

35. D

Sol. Ph_3C-X, has great tendency to undergo S_N1 and S_N2 both, due to very good stability of carbocation in S_N1 reaction mechanism and very good stability of transition state under S_N2 conditions, respectively.

36. B

Sol. Pinacole-Pinacolone reaction occurs but free carbocation is not formed under the given conditions. Hence, NGP by phenyl ring results into the product.



37. C

Sol. $2CHCl_3 + O_2 \xrightarrow{h\nu} 2COCl_2 + 2HCl$;

38. C

Sol. (A) Clean water would have BOD value of less than 5 ppm whereas highly polluted water would have a BOD value of 17 ppm or more.

(B) High BOD means high activity of bacteria.

(D) Synthetic chlorinated pesticides are non-biodegradable.

39. A

Sol. Among the given four

Only Ranitidine is sulphur containing compound. However, all of these contains N as essential element.

40. D

Sol. Following complexes can be derived from the given formula

$\left[Cr(H_2O)_6\right]BrCl_2$; $\left[CrBr(H_2O)_5\right]Cl_2.H_2O$;

$\left[CrCl(H_2O)_5\right]ClBr.H_2O$; $\left[CrCl_2(H_2O)_4\right]Br.2H_2O$;

$\left[CrBrCl(H_2O)_4\right]Cl.2H_2O$ and none of those can exhibit optical isomerism.

41. D

42. C

Sol. $Zn + dil.HNO_3 \longrightarrow Zn(NO_3)_2 + N_2O\uparrow$;

N_2O is neutral and doesn't form brown ring with $FeSO_4$ solution.

43. B

Sol. All others are amphoteric in nature, hence soluble in aq. NaOH.

44. D

Sol. $BiCl_3$ undergo partial hydrolysis to give BiOCl (pearl white), which appears as white turbidity.

45. C

Sol.

$$\underset{(P)}{Hg_2Cl_2} + NH_3 \longrightarrow \underset{(Q)(Black)}{HgNH_2Cl.Hg}$$

$$\underset{(Q)}{HgNH_2Cl.Hg} \xrightarrow{HNO_3\ /\ HCl} \underset{(R)}{HgCl_2}$$

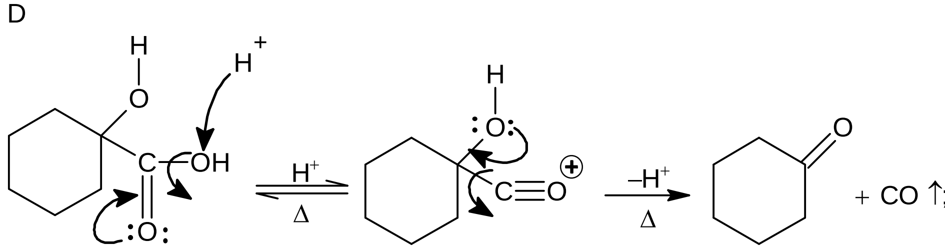
$$\underset{(S)}{K_2HgI_4} \xleftarrow{\text{excess KI}} \underset{(Orange)}{HgI_2} \xleftarrow{KI} \underset{(R)}{HgCl_2}$$

46. B

Sol. Primary amines give carbyl amine test; to produce foul smelling R – NC (isocyanides).

47. D

Sol.



48. C

Sol. The 'CHO' group, para to 'NO_2' is more electrophilic hence it is attacked by OH^- and oxidised and the other CHO (meta to NO_2) is reduced due to intramolecular Cannizzaro reaction.

49. C

Sol. Oxide ions constitute ccp means fcc unit cells, i.e. Number of oxide ions per unit cell = 4

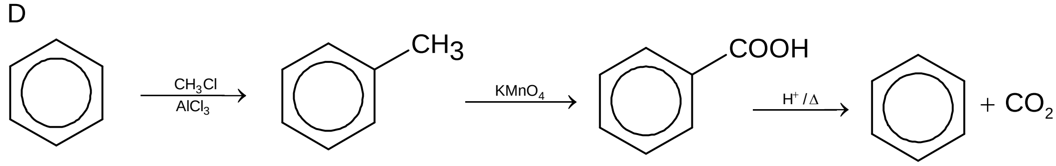
Number of 'A' per unit cell $= \frac{4}{2} = 2$ (As 'A' occupy half of the octahedral voids)

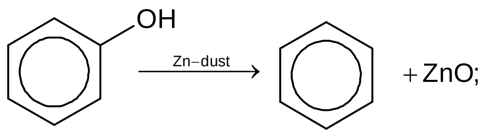
Number of 'B', per unit cell $= \frac{12.5}{100} \times 8 = 1$

Hence, formula becomes A_2BO_4.

50. D

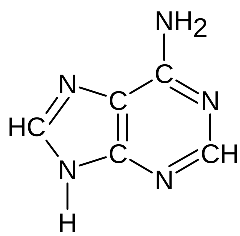
Sol.





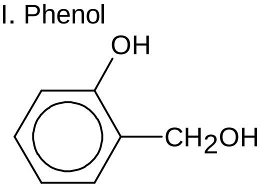
SECTION – B

51. 00005.00
Sol.

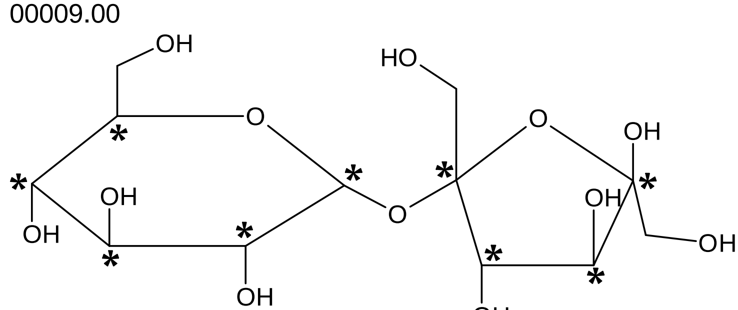


(Adenine)

52. 00006.00
Sol.

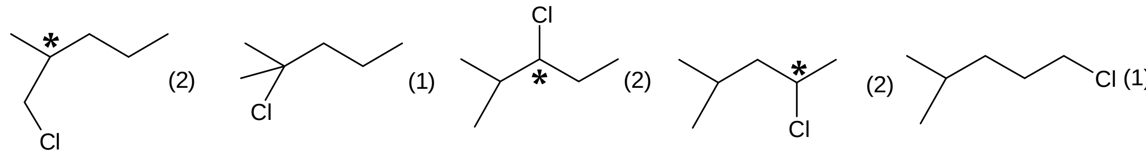
Polymer	**Monomer(s)**	
Buna-S	I. Styrene	II. 1,3-Butadiene
Teflon	$CF_2 = CF_2$	
Cellulose	Glucose	
Dacron	I. Ethylene glycol	II. Terephthalic acid
Nylon 6,6	I. Adipic acid	II. $H_2N(CH_2)_6NH_2$
Bakelite	I. Phenol	II. HCHO
Novolac		
Neoprene	2-chloro-1,3-butadiene	
Buna-N	I. 1,3-butadiene	II. Acrylonitrile
Glyptal	I. Ethylene glycol	II. Phthalic acid

53. 00009.00
Sol.



54. 00008.00

Sol.



55. 00001.00

Sol. $\Delta T_f = K_f m; (i = 1)$

$\Delta T_f = 1.86 \times m$

$\Delta T_b = K_b m; (i = 1)$

= 0.52 m

For said difference of melting point and boiling point,

$\Delta T_f + \Delta T_b = 4.76$ (in case of water)

1.86 m + 0.52 m = 4.76

m = 2.0

Mass of solvent =0.5 kg

Hence, n = 1.00 mole

56. 00009.00

Sol. For a zero order reaction

$t_{1/2} = \frac{a}{2k}$ or $k = \frac{a}{2t_{1/2}}$

and x = kt

$\frac{3a}{4} = k.t_{75\%}$

$t_{75\%} = \frac{3a}{4 \times k} = \frac{3a}{4 \times \frac{a}{2t_{1/2}}} = \frac{6t_{1/2}}{4}$

$= \frac{6 \times 6}{4} = 9 \text{ min}$.

57. 00003.00

Sol. 20 ml gold sol require 36 mg sodium oleate hence 10 ml of given gold sol require 18 mg sodium oleate to protect from coagulation by 3 ml, 20% NaCl (aq). Hence, these 10 ml sol would require $\frac{18}{3 \times 2} = 3 \text{ mg}$. Sodium oleate to protect from coagulation by 1 ml, 10% NaCl (aq).

58. 00002.00

Sol. Let, overall rate constant for consumption of 'A' is k then $k = k_1 + k_2 = 3k_1$.

Hence, half-life of $A = \frac{0.693}{k}$

$= \frac{0.693}{3k_1} = \frac{0.693}{3 \times 0.0693} = \frac{10}{3} \text{min}$

Number of half-lives passed till 10 min = 3

Hence, $P_{A,t} = 16 \times \left(\frac{1}{2}\right)^3 = 2 \text{ atm}$.

59. 00003.00

Sol. $E^o_{cell} = 0.534 - 0.5196 = 0.0144$

$E^o_{cell} = \frac{0.06}{2}\log K_{eq}$

$\frac{0.0144 \times 2}{0.06} = \log K_{eq}$

$0.48 = \log K_{eq}$

$K_{eq} = 3$

60. 00004.00

Sol. $W = +\pi ab = 3.14 \times \frac{1}{2} \times \frac{5.1}{2}$

= 4 (atm-litre)

Mathematics

PART – C

SECTION – A

61. C

Sol. $\lim_{m\to\infty}(a_m)^{\frac{1}{m}} = \left(\left(\left(1+\frac{1}{m}\right)^{m+1} - \frac{m+1}{m}\right)^{-m}\right)^{\frac{1}{m}} = \frac{1}{e-1}$

62. A

Sol. $f(x^2 + 1) = (x^2 + 1)^2 + 3(x^2 + 1) - 2$

$\Rightarrow f(x) = x^2 + 3x - 2$ for all $x \geq 1$

$\int f(x) = \frac{x^3}{3} + \frac{3x^2}{2} - 2x + c$

63. A

Sol. $\frac{\int_0^1 x^{\alpha}}{\int_0^1 x^{\beta}} = \frac{\frac{1}{\alpha+1}}{\frac{1}{\beta+1}} = \frac{\beta+1}{\alpha+1}$

64. D

Sol. Let the equation of circle is $(x - 15)^2 + (y + 3)^2 = r^2$

Solving circle with parabola $(x - 15)^2 + \left(\frac{x^2}{3}+3\right)^2 = r^2$

Now, apply the condition of coincident roots to find r (radius)

65. C

Sol. Since f(x) is increasing function. Hence, $f(x) - 2x^3 \geq 2x^3 - f(x)$

66. C

Sol. Let $x_1 < x_2$

$\Rightarrow 3^{\frac{1}{7}} + 9^{\frac{1}{9}} < 5^{\frac{1}{7}} + 7^{\frac{1}{7}} \Rightarrow 9^{\frac{1}{7}} - 7^{\frac{1}{7}} < 5^{\frac{1}{7}} - 3^{\frac{1}{7}}$

$\Rightarrow (1+8)^{\frac{1}{7}} - (8-1)^{\frac{1}{7}} < (4+1)^{\frac{1}{7}} - (4-1)^{\frac{1}{7}}$

$\Rightarrow (1+8)^{\frac{1}{7}} + (1-8)^{\frac{1}{7}} < (1+4)^{\frac{1}{7}} + (1-4)^{\frac{1}{7}}$

Now, apply $a^7 + b^7 = (a + b)(a^6 - a^5b + a^4b^2 - a^3b^3 + a^2b^4 - ab^5 + b^6)$

67. D

Sol. $\begin{vmatrix} 1 & 1 & 1 \\ c+2 & c+4 & 6 \\ (c+2)^2 & (c+4)^2 & 36 \end{vmatrix} = 0$

68. C

Sol. $a = \log_{12} 18 = \frac{\log_e 18}{\log_e 12}$, $b = \frac{\log_e 54}{\log_e 24}$, put the values of a and b

69. D

Sol. Let $\overline{OA} = \vec{a}$, $\overline{OB} = \vec{b}$, $\overline{OC} = \vec{c}$

$\bar{a}\,.\,\bar{a} + (\bar{b} - \bar{c}) \cdot (\bar{b} - \bar{c}) = \bar{b} \cdot \bar{b} + (\bar{c} - \bar{a}) \cdot (\bar{c} - \bar{a})$

$\Rightarrow 2\bar{b} \cdot \bar{c} = 2\bar{c} \cdot \bar{a} \Rightarrow (\bar{a} - \bar{b}) \cdot \bar{c} = 0$

$\Rightarrow AB \perp OC$ similarly $\overline{BC} \perp \overline{OA}$ and $\overline{CA} \perp \overline{OB}$

70. C

Sol. $\dfrac{1}{SP} + \dfrac{1}{SQ} = \dfrac{2}{2a}$

71. A

Sol. Let equation of normal at P(a cos θ, b sin θ) is ax sec θ – by cosec θ = $a^2 - b^2$

Coordinate of G is $\left(\dfrac{a^2 - b^2}{a}\cos\theta, 0\right)$; $PG = \dfrac{b}{a}\sqrt{(a^2 \sin^2\theta + b^2\cos^2\theta)}$

$PF = \dfrac{ab}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}$

72. A

Sol. $\Delta_{DEF} = \dfrac{1}{2}(a\cos A)(b\cos B)\sin(\pi - 2c)$; $\Delta ABC = \dfrac{1}{2}ab\sin C$

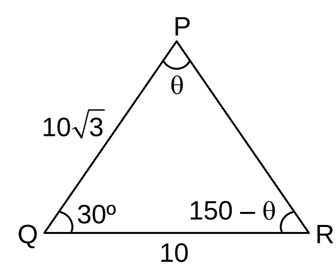
73. A

Sol. (A) $\dfrac{10}{\sin\theta} = \dfrac{10\sqrt{3}}{\sin(150° - \theta)} \Rightarrow \theta = 30°$

(B) $\Delta = \dfrac{1}{2}(10)10\sqrt{3}\sin 30° = 25\sqrt{3}$

(C) $r = \dfrac{\Delta}{S} = 10\sqrt{3} - 15$

(D) area = πR^2 = $\dfrac{\pi}{4}\left(\dfrac{10}{1/2}\right)^2 = 100\pi$



74. C

Sol. $\begin{bmatrix} d_1 & 0 & 0 & 0 & 0 & .... \\ 0 & d_2 & 0 & 0 & 0 & ..... \\ 0 & 0 & d_3 & 0 & .... & ..... \\ ..... & ..... & ..... & ..... & ..... & ..... \\ ..... & ..... & ..... & ..... & ..... & ..... \end{bmatrix} \begin{bmatrix} \dfrac{1}{d_1} & 0 & 0 & 0 & ..... \\ 0 & \dfrac{1}{d_2} & 0 & 0 & ..... \\ 0 & 0 & \dfrac{d}{d_3} & 0 & ..... \\ ..... & ..... & ..... & ..... & ..... \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & ..... \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ ..... & ..... & ..... & ..... \end{bmatrix}$

75. A

Sol. cos A + cos C = 2 – 2 cos B

$\Rightarrow 2\cos\left(\dfrac{A+C}{2}\right)\cos\left(\dfrac{A-C}{2}\right) = 2 \cdot 2\sin^2\left(\dfrac{B}{2}\right)$; $\cos\dfrac{A+C}{2} = 2\sin\dfrac{B}{2} = 2\cos\left(\dfrac{A+C}{2}\right)$

$\Rightarrow \cos\dfrac{A}{2}\cos\dfrac{C}{2} + \sin\dfrac{A}{2}\sin\dfrac{C}{2} = 2\left(\cos\dfrac{A}{2}\cos\dfrac{C}{2} - \sin\dfrac{A}{2}\sin\dfrac{C}{2}\right)$

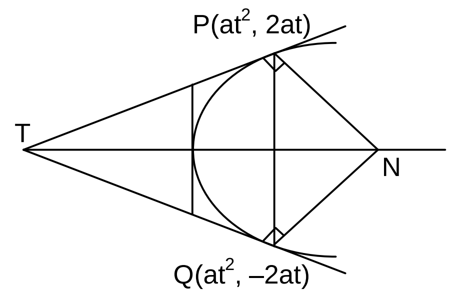
$\Rightarrow \tan\frac{A}{2}\tan\frac{C}{2} = \frac{1}{3} \Rightarrow \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}\sqrt{\frac{(s-a)(s-b)}{s(s-c)}} = \frac{1}{3}$

$\Rightarrow$ 3s – 3b = s a + b + c = 3b $\Rightarrow$ a + c = 2b

76. D

Sol. $\angle TPN = \angle TQN = 90°$

Hence, PTQN is a cyclic quadrilateral



77. B

Sol. $\int_0^1 f(x)dx + \int_0^1 f(1+x)dx + \int_0^1 f(2+x)dx + ...\int_0^1 f(99+x)dx$

$\int_0^1 f(x)dx + \int_1^2 f(x)dx + \int_2^3 f(x)dx + ... \int_{99}^{100} f(x)dx = \int_0^{100} f(x)dx = a$

78. A

Sol. Solution equation of common chord is $S_1 - S_2 = 0$ where $S_1 = x^2 + y^2 + 8x + 4y - 5$

$S_2 = x^2 + y^2 + \lambda(x - y) = 0$

79. B

Sol. $p'(0) = 0 \Rightarrow c = 0$

$p'(x) = 0$ has only one real root the $p(x) = 0$ has maximum two real roots

80. D

Sol. $(\bar{a}\cdot\bar{c})\bar{b} - (\bar{a}\cdot\bar{b})\bar{c} = \frac{\bar{b}+\bar{c}}{\sqrt{2}} \Rightarrow \bar{a}\cdot\bar{b} = -\frac{1}{\sqrt{2}} \Rightarrow \cos\theta = -\frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{3\pi}{4}$

SECTION – B

81. 00003.00

Sol. $f(g(x)) = x \Rightarrow f'(g(x))g'(x) = 1$

$\frac{d}{dx}\left(\frac{g(x)}{g(g(x))}\right) = \frac{g(g(x))g'(x) - g'(g(x))g'(x)g(x)}{(g(g(x)))^2}$

Also, $g(4) = 0$ and $g'(4) = \frac{1}{f'(0)}$

$f'(x) = 3x^2 + 3 \Rightarrow f'(0) = 3$, $g(g(4)) = g(0) = -1$

82. 00003.00

Sol. $\sin^4 x + 2\sin^2 x + 1 - 3 + a^2 = 0$

$a^2 = 3 - (1 + \sin^2 x)^2 \Rightarrow 0 \le a^2 \le 2 \Rightarrow a = 0, \pm 1$

83. 00007.00

Sol. $v = \begin{vmatrix} a_1 & a_2 & a_3 \\ a_2 & a_3 & a_1 \\ a_3 & a_1 & a_2 \end{vmatrix} = \left(\dfrac{a_1+a_2+a_3}{2}\right)\left((a_1-a_2)^2+(a_2-a_3)^2+(a_3-a_1)^2\right) = 108$

$\dfrac{3a_2}{2}[6d^2] = 108 \Rightarrow a_2 d^2 = 4 \times 3 \Rightarrow d = 2,\ a_2 = 3$

84. 00000.00

Sol. RHS = $(x - 2)^2 + 1$

$\Rightarrow$ Minimum value of RHS is 1 at x = 2 but maximum value of LHS is 1 for some odd number

85. 00006.00

Sol. $P\left(\dfrac{A}{B}\right) = P(A) = \dfrac{2}{3} \Rightarrow P(\bar{A}) = \dfrac{1}{3} \Rightarrow P(B) = 1 - \dfrac{2}{3} = \dfrac{1}{3}$

$P(\bar{A} \cap B) = P(\bar{A})P(B) = \dfrac{1}{3} \times \dfrac{1}{3} = \dfrac{1}{9}$

86. 00002.00

Sol. Vector $\hat{j}$ and $(2\hat{i}+2\hat{j}-2\hat{k})$ both are parallel to plane, then the equation of plane is

$(\vec{r} - (\hat{i}+\hat{k})) \cdot (\hat{j} \times (\hat{i}+\hat{j}-\hat{k})) = 0$

$\Rightarrow ((x-1)\hat{i} + y\hat{j} + (z-1)\hat{k}) \cdot (-\hat{k} - \hat{i}) = 0$

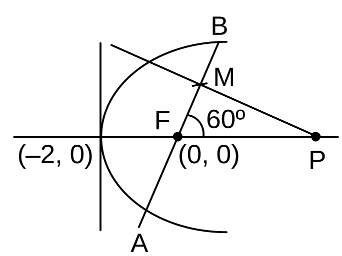
$\Rightarrow x + z - 2 = 0 \Rightarrow |a + b + c| = 2$

87. 00004.00

Sol. Length of focal chord $AB = \dfrac{32}{3}$

MF = AM – AF = $\dfrac{1}{2}AB - \dfrac{8}{3} = \dfrac{8}{3}$

PF = MF sec 60° = $\dfrac{16}{3}$



88. 00002.00

Sol. $\dfrac{z_1-1}{z_1+1} = \left(\dfrac{z_2^2+1-2z_2}{z_2^2+1+2z_2}\right) = \left(\dfrac{\dfrac{z_2^2+1}{2z_2}-1}{\dfrac{z_2^2+1}{2z_2}+1}\right) \Rightarrow z_1 = \dfrac{1}{2}\left(z_2 + \dfrac{1}{z_2}\right)$

Let $z_2 = \cos\theta + i\sin\theta \Rightarrow z_1 = \cos\theta$

$\Rightarrow |z_1 - 1| + |z_1 + 1| = |\cos\theta - 1| + |\cos\theta + 1| = \left|2\sin^2\dfrac{\theta}{2}\right| + \left|2\cos^2\dfrac{\theta}{2}\right| = 2$

89. 00009.00

Sol. If all wrong then 44 ways, for one right and four wrong is 5 × 9 = 45. If 2 right and three wrong then ${}^5C_2(2) = 20$. If 3 right and two wrong then ${}^5C_3(1) = 10$

90. 00004.00

Sol. $\int_0^{\infty} x^9 e^{-x^2} dx = \int_0^{\infty} x^8 \left(xe^{-x^2}\right) dx$. Put $x^2 = -t \Rightarrow 2x\, dx = -dt$

$\Rightarrow I_1 = -\frac{1}{2}\int_0^{-\infty} t^4 e^t dt = \frac{1}{2}(4!)$. Similarly, $I_2 = (3!)$

$\Rightarrow \frac{I_1}{I_2} = 4$